RELATIVE WEAKLY ASSOCIATED POINTS UNDER A FINITE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The proof is a fibre-wise reduction; the absolute statement it invokes is itself a card of this pool (L44219), and it enters the assembly as a navigation pointer.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Base change of a pushforward along an affine morphism. Cohomology of Schemes, Lemma coherent-lemma-affine-base-change, used at proof lines 1234-1237. Identifies the fibre of i_*F with the pushforward of the fibre.
- Weakly associated points under a finite morphism. Lemma lemma-weakly-associated-finite (879-884, produced as L44219), used at proof lines 1237-1239. Is the absolute statement applied on each fibre.

Source reference. Original lemma 02KG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3371–3371. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05EZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7008–7008. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Let $i : Z \to X$ be a finite morphism. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_Z$-module. Then $\mathit{WeakAss}_{X/S}(i_*\mathcal{F}) = i(\mathit{WeakAss}_{Z/S}(\mathcal{F}))$.*

Proof. Let $i_s : Z_s \to X_s$ be the induced morphism between fibres. Then $(i_*\mathcal{F})_s = i_{s,*}(\mathcal{F}_s)$ by Cohomology of Schemes, Lemma 02KG and the fact that i is affine. Hence we may apply Lemma 05EZ to conclude. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.